COMPUTER ARCHITECTURE

BRAC UNIVERSITY
Inspiring Excellence

Understanding Performance
(PBK HandOut Edition)

Hardware/Software Component	How this component affects performance?	Where to learn?
Algorithm/Source-Level Statements	The algorithm here simply refers to the original program itself. *The more optimized the algorithm the better the performance.* ***Determines the #operations & complexity of the program / functions.***	cse221
Programming Language, Compiler & architecture	The software tools used to write and convert the program into machine instructions. ***Determines the #machine-instructions to be executed for the program written / for each function / statement.***	Chapter 2 & 3
Processor + Memory system	Determines how fast instructions can be executed	Chapter 4, 5, 6
I/O system + OS	How efficiently the system can handle the input and output operations.	Chapter 4, 5, 6

Software

Hardware

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SYSTEM STACK OVERVIEW

Performance Factors Across the Stack

Algorithm

Dictates total source statements and execution volume of I/O operations.

CSE221

Language & Compiler

Determines machine instructions produced per statement.

Chapters 2 & 3

Processor & Memory

Determines execution cycle throughput and memory latency.

Chapters 4, 5 & 6

I/O & Operating System

Dictates physical throughput and interrupt/driver overhead.

Chapters 4, 5 & 6